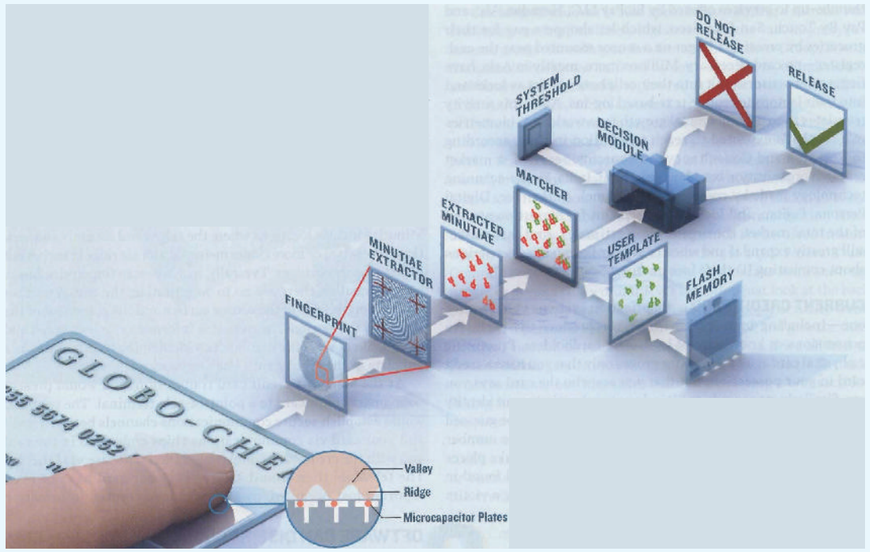


Figure TF9-6 Elements of a fingerprint matching system. (Courtesy of IEEE Spectrum.)

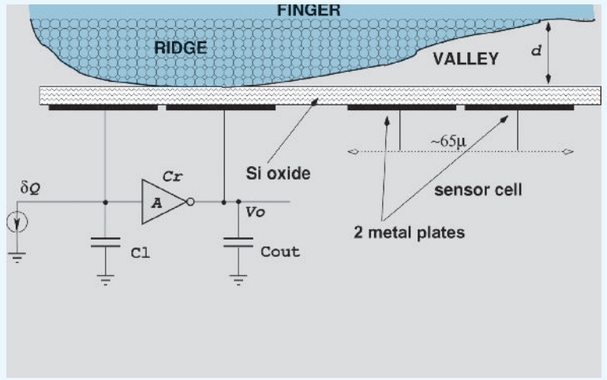


Figure TF9-7 Fingerprint representation.

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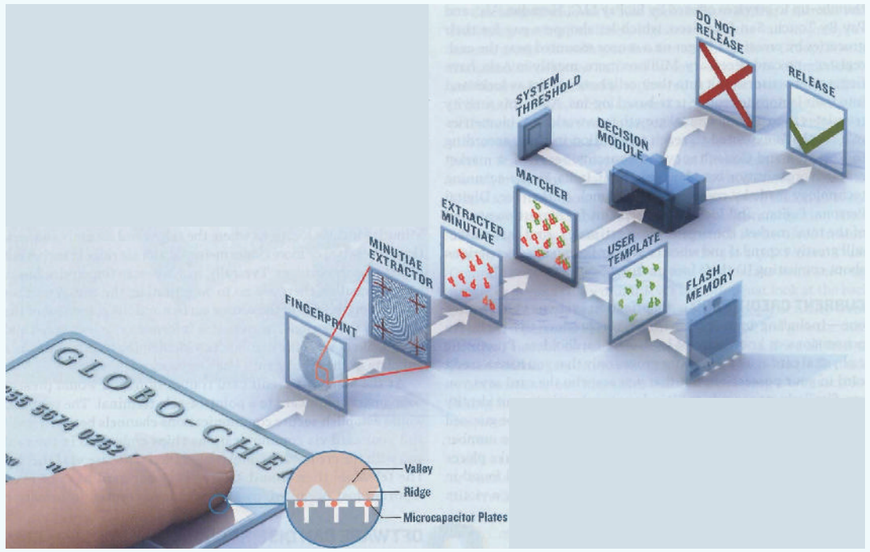


图 TF9-6 指纹匹配系统的要素。（由 IEEE Spectrum 提供。）

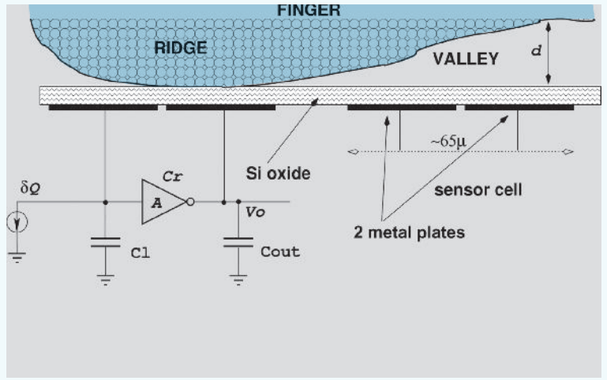


图 TF9-7 指纹表示。

Exercise 4-16: Find $\mathbf{E}_1$ in **Fig. 4-21** if

$$\mathbf{E}_2 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3 \text{ (V/m)},$$
$$\varepsilon_1 = 2\varepsilon_0,$$
$$\varepsilon_2 = 8\varepsilon_0,$$

and the boundary is charge-free.

Answer: $\mathbf{E}_1 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}12$ (V/m). (See EM.)

Exercise 4-17: Repeat Exercise 4.16 for a boundary with surface charge density $\rho_s = 3.54 \times 10^{-11}$ (C/m^2).

Answer: $\mathbf{E}_1 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}14$ (V/m). (See EM.)

4-8.1 Dielectric-Conductor Boundary

Consider the case when medium 1 is a dielectric and medium 2 is a perfect conductor. In a perfect conductor, because electric fields and fluxes vanish, it follows that $\mathbf{E}_2 = \mathbf{D}_2 = 0$, which implies that components of $\mathbf{E}_2$ and $\mathbf{D}_2$ tangential and normal to the interface are zero. Consequently, from Eq. (4.100) and Eq. (4.104), the fields in the dielectric medium at the boundary with the conductor satisfy

$$E_{1t} = D_{1t} = 0, \tag{4.110a}$$
$$D_{1n} = \varepsilon_1 E_{1n} = \rho_s. \tag{4.110b}$$

These two boundary conditions can be combined into

$$\mathbf{D}_1 = \varepsilon_1 \mathbf{E}_1 = \hat{\mathbf{n}}\rho_s, \quad \text{(at conductor surface)} \tag{4.111}$$

where $\hat{\mathbf{n}}$ is a unit vector directed normally outward from the conducting surface.

► The electric field lines point directly away from the conductor surface when ρ_s is positive and directly toward the conductor surface when ρ_s is negative. ◄

Figure 4-22 shows an infinitely long conducting slab placed in a uniform electric field $\mathbf{E}_1$. The media above and below the slab have permittivity ε_1. Because $\mathbf{E}_1$ points away from the upper surface, it induces a positive charge density $\rho_s = \varepsilon_1 |\mathbf{E}_1|$ on the upper slab surface. On the bottom surface, $\mathbf{E}_1$ points toward the surface; therefore, the induced charge density is $-\rho_s$. The presence of these surface charges induces an electric field $\mathbf{E}_i$ in the conductor, resulting in a total field $\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_i$. To satisfy the condition that $\mathbf{E}$ must be everywhere zero in the conductor, $\mathbf{E}_i$ must equal $-\mathbf{E}_1$.

If we place a metallic sphere in an electrostatic field (**Fig. 4-23**), positive and negative charges accumulate on the upper and lower hemispheres, respectively. The presence of the sphere causes the field lines to bend to satisfy the condition expressed by Eq. (4.111); that is, ***E is always normal to a conductor boundary.***

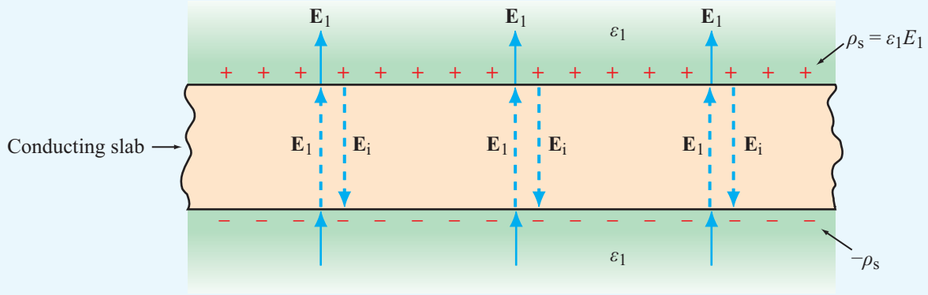


Figure 4-22 When a conducting slab is placed in an external electric field $\mathbf{E}_1$, charges that accumulate on the conductor surfaces induce an internal electric field $\mathbf{E}_i = -\mathbf{E}_1$. Consequently, the total field inside the conductor is zero.

练习 4-16：在图 4-21 中求 E_1，如果

$$\mathbf{E}_2 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}3 \text{ (V/m)},$$
$$\varepsilon_1 = 2\varepsilon_0,$$
$$\varepsilon_2 = 8\varepsilon_0,$$

并且界面上无自由面电荷（$\rho_s = 0$）。

Answer: $\mathbf{E}_1 = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}12$ (V/m). (See EM.)

练习 4-17：对表面电荷密度 $\rho_s = 3.54 \times 10^{-11}$ (C/m^2) 的边界重复练习 4.16。

答案：$E_1 = \hat{x}2 - \hat{y}3 + \hat{z}14$ (V/m)。 （参见EM。）

4-8.1 电介质-导体边界

考虑介质 1 是电介质且介质 2 是理想导体的情况。在理想导体中，由于电场和通量消失，因此 $E_2 = D_2 = 0$，这意味着 E_2 和 D_2 与界面相切和法向的分量为零。因此，从方程。 (4.100) 和等式。 (4.104)，介电介质中与导体边界处的场满足

$$E_{1t} = D_{1t} = 0, \tag{4.110a}$$
$$D_{1n} = \varepsilon_1 E_{1n} = \rho_s. \tag{4.110b}$$

这两个边界条件可以合并为

$$\mathbf{D}_1 = \varepsilon_1 \mathbf{E}_1 = \hat{\mathbf{n}}\rho_s, \quad \text{(at conductor surface)} \tag{4.111}$$

其中 $\hat{n}$ 是从导电表面法向向外的单位矢量。

► 当 ρ_s 为正时，电场线直接远离导体表面；当 ρ_s 为负时，电场线直接指向导体表面。 ◄

图 4-22 显示了放置在均匀电场 E_1 中的无限长导电板。板上方和下方的介质具有介电常数 ε_1。由于 E_1 指向远离上表面的方向，因此它会在上板表面上感应出正电荷密度 $\rho_s = \varepsilon_1 |E_1|$。在底面上，$E_1$指向曲面；因此，感应电荷密度为$-\rho_s$。这些表面电荷的存在会在导体中感应出电场 E_i，从而产生总电场 $E = E_1 + E_i$。为了满足导体中 E 必须处处为零的条件，E_i 必须等于 $-E_1$。

如果我们将一个金属球放在静电场中（图 4-23），正电荷和负电荷分别积聚在上半球和下半球上。球体的存在导致场线弯曲以满足方程（1）表示的条件。 (4.111)；也就是说，E 始终垂直于导体边界。

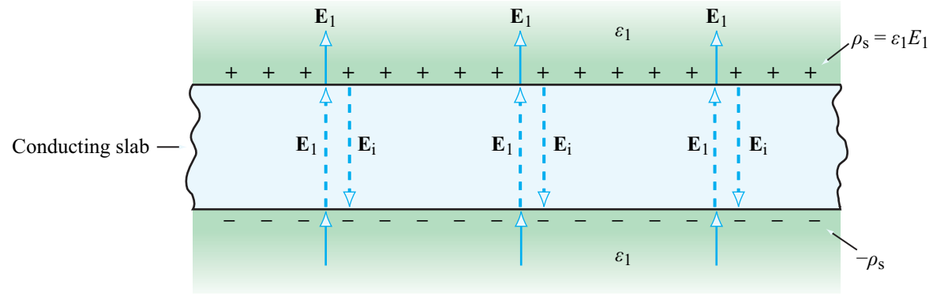
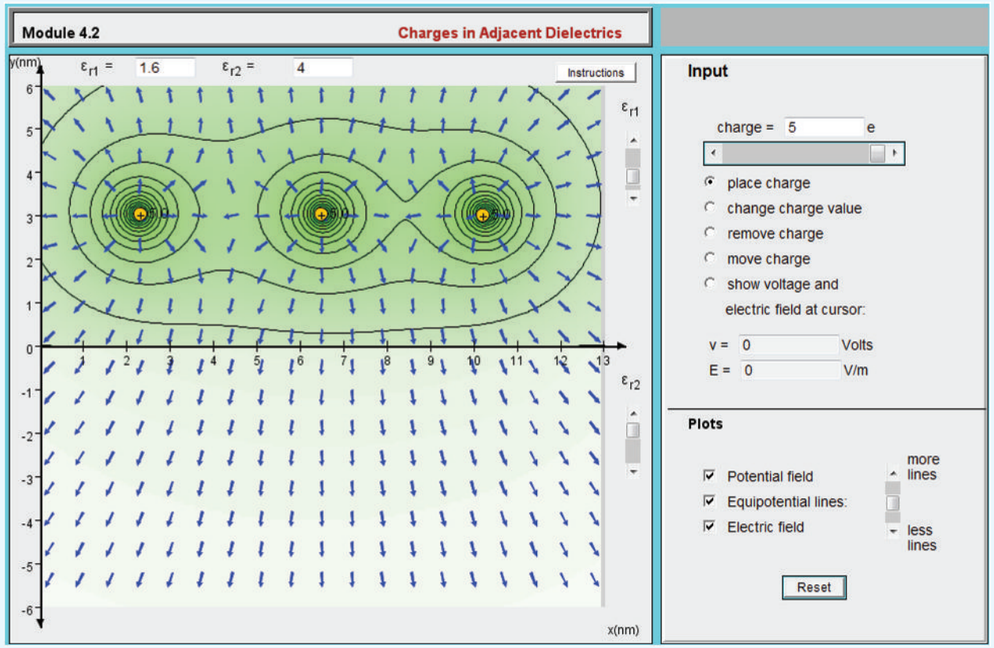


图 4-22 当导电板置于外部电场 E_1 中时，积聚在导体表面的电荷会感应出内部电场 $E_i = -E_1$。因此，导体内的总电场为零。

Module 4.2 Charges in Adjacent Dielectrics In two adjoining half-planes with selectable permittivities, the user can place point charges anywhere in space and select their magnitudes and polarities. The module then displays **E**, V, and the equipotential contours of V.



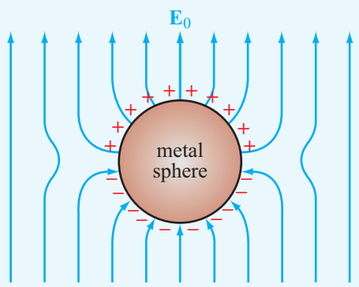


Figure 4-23 Metal sphere placed in an external electric field $\mathbf{E}_0$.

4-8.2 Conductor–Conductor Boundary

We now examine the general case of the boundary between two media—neither of which is a perfect dielectric or a perfect conductor (**Fig. 4-24**). Medium 1 has permittivity ε_1 and conductivity σ_1, medium 2 has ε_2 and σ_2, and the interface between them holds a surface charge density ρ_s. For the electric fields, Eqs. (4.100) and (4.105b) give

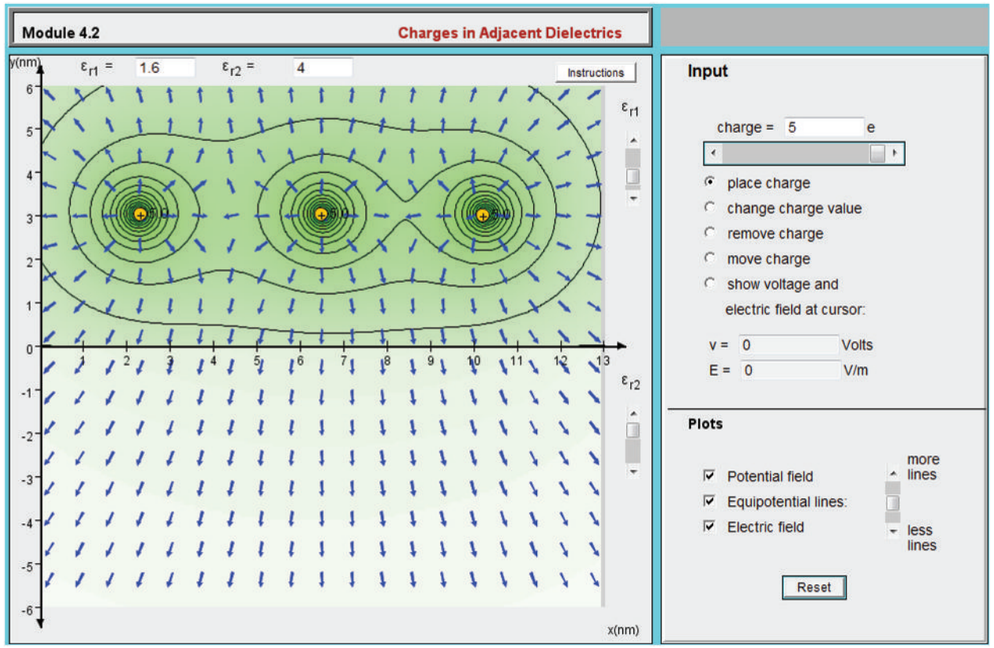
$$\mathbf{E}_{1t} = \mathbf{E}_{2t}, \qquad \varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \rho_s. \tag{4.112}$$

Since we are dealing with conducting media, the electric fields give rise to current densities $\mathbf{J}_1 = \sigma_1 \mathbf{E}_1$ and $\mathbf{J}_2 = \sigma_2 \mathbf{E}_2$. Hence,

$$\frac{\mathbf{J}_{1t}}{\sigma_1} = \frac{\mathbf{J}_{2t}}{\sigma_2}, \qquad \varepsilon_1 \frac{J_{1n}}{\sigma_1} - \varepsilon_2 \frac{J_{2n}}{\sigma_2} = \rho_s. \tag{4.113}$$

The tangential current components $\mathbf{J}_{1t}$ and $\mathbf{J}_{2t}$ represent currents flowing in the two media in a direction parallel to the

模块 4.2 相邻电介质中的电荷 在具有可选介电常数的两个相邻半平面中，用户可以将点电荷放置在空间中的任何位置，并选择它们的大小和极性。然后模块显示 E、V 和 V 的等势线。



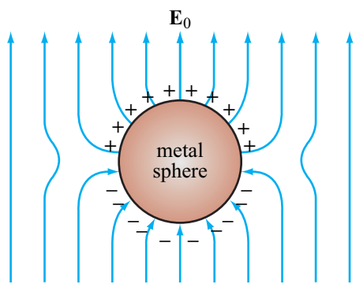


图 4-23 放置在外部电场 E_0 中的金属球。

4-8.2 导体-导体边界

我们现在研究两种介质之间边界的一般情况——这两种介质都不是理想的电介质或理想的导体（图4-24）。介质1具有介电常数ε_1和电导率σ_1，介质2具有ε_2和σ_2，并且它们之间的界面具有表面电荷密度ρ_s。对于电场，方程。 (4.100) 和 (4.105b) 给出

$$\mathbf{E}_{1t} = \mathbf{E}_{2t}, \qquad \varepsilon_1 E_{1n} - \varepsilon_2 E_{2n} = \rho_s. \tag{4.112}$$

由于我们处理的是导电介质，电场会产生电流密度 $J_1 = \sigma_1 E_1$ 和 $J_2 = \sigma_2 E_2$。因此，

$$\frac{\mathbf{J}_{1t}}{\sigma_1} = \frac{\mathbf{J}_{2t}}{\sigma_2}, \qquad \varepsilon_1 \frac{J_{1n}}{\sigma_1} - \varepsilon_2 \frac{J_{2n}}{\sigma_2} = \rho_s. \tag{4.113}$$

切向电流分量 J_{1t} 和 J_{2t} 表示在两种介质中沿平行于介质方向流动的电流。

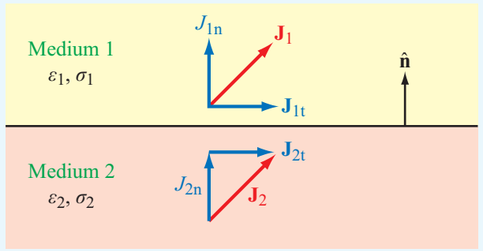


Figure 4-24 Boundary between two conducting media.

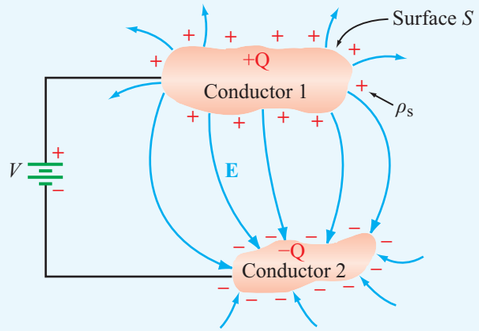


Figure 4-25 A dc voltage source connected to a capacitor composed of two conducting bodies.

boundary; hence, there is no transfer of charge between them. This is not the case for the normal components. If $J_{1n} \neq J_{2n}$, then a different amount of charge arrives at the boundary than leaves it. Hence, ρ_s cannot remain constant in time, which violates the condition of electrostatics requiring all fields and charges to remain constant. Consequently, *the normal component of* $\mathbf{J}$ *has to be continuous across the boundary between two different media under electrostatic conditions.* Upon setting $J_{1n} = J_{2n}$ in Eq. (4.113), we have

$$J_{1n}\left(\frac{\varepsilon_1}{\sigma_1} - \frac{\varepsilon_2}{\sigma_2}\right) = \rho_s \quad \textbf{(electrostatics)}. \tag{4.114}$$

Concept Question 4-22: What are the boundary conditions for the electric field at a conductor–dielectric boundary?

Concept Question 4-23: Under electrostatic conditions, we require $J_{1n} = J_{2n}$ at the boundary between two conductors. Why?

4-9 Capacitance

When separated by an insulating (dielectric) medium, any two conducting bodies, regardless of their shapes and sizes, form a ***capacitor***. If a dc voltage source is connected across them (**Fig. 4-25**) the surfaces of the conductors connected to the positive and negative source terminals accumulate charges $+Q$ and $-Q$, respectively.

► When a conductor has excess charge, it distributes the charge on its surface in such a manner as to maintain a zero electric field everywhere within the conductor, thereby ensuring that the electric potential is the same at every point in the conductor. ◄

The ***capacitance*** of a two-conductor configuration is defined as

$$C = \frac{Q}{V} \quad \text{(C/V or F)}, \tag{4.115}$$

where V is the potential (voltage) difference between the conductors. Capacitance is measured in farads (F), which is equivalent to coulombs per volt (C/V).

The presence of free charges on the conductors' surfaces gives rise to an electric field $\mathbf{E}$ (**Fig. 4-25**) with field lines originating on the positive charges and terminating on the negative ones. Since the tangential component of $\mathbf{E}$ always vanishes at a conductor's surface, $\mathbf{E}$ is always perpendicular to the conducting surfaces. The normal component of $\mathbf{E}$ at any point on the surface of either conductor is given by

$$E_n = \hat{\mathbf{n}} \cdot \mathbf{E} = \frac{\rho_s}{\varepsilon}, \quad \textbf{(at conductor surface)} \tag{4.116}$$

where ρ_s is the surface charge density at that point, $\hat{\mathbf{n}}$ is the outward normal unit vector at the same location, and ε is the permittivity of the dielectric medium separating the

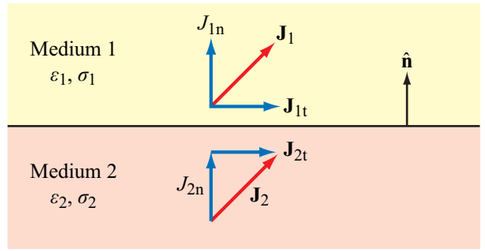


图 4-24 两种导电介质之间的边界。

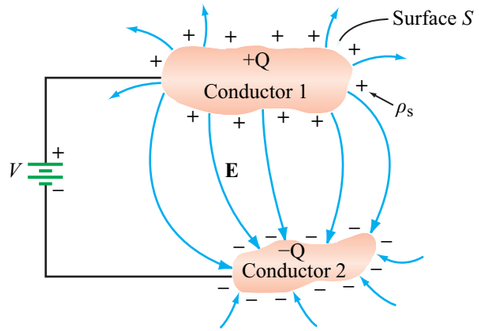


图 4-25 连接到由两个导电体组成的电容器的直流电压源。

边界;因此，它们之间不存在电荷转移。对于普通组件来说，情况并非如此。如果 $J_{1n} \neq J_{2n}$，则到达边界的电荷量与离开边界的电荷量不同。因此，ρ_s 不能随时间保持恒定，这违反了静电学要求所有场和电荷保持恒定的条件。因此，在静电条件下，J 的法向分量必须连续穿过两种不同介质之间的边界。在等式中设置 $J_{1n} = J_{2n}$ 后（4.113），我们有

$$J_{1n}\left(\frac{\varepsilon_1}{\sigma_1} - \frac{\varepsilon_2}{\sigma_2}\right) = \rho_s \quad \textbf{(electrostatics)}. \tag{4.114}$$

概念问题 4-22：导体-电介质边界电场的边界条件是什么?

概念问题 4-23：
在静电条件下，我们要求两个导体之间的边界处有 $J_{1n} = J_{2n}$。为什么?

4-9 电容

当被绝缘（介电）介质隔开时，任何两个导电体，无论其形状和大小如何，都会形成电容器。如果直流电压源跨接在它们之间（图 4-25），则连接到正源极端子和负源极端子的导体表面将分别积聚电荷 +Q 和 −Q。

► 当导体有多余的电荷时，它会将电荷分布在其表面，以保持导体内各处的电场为零，从而确保导体内每一点的电势相同。 ◄

双导体配置的电容定义为

$$C = \frac{Q}{V} \quad \text{(C/V or F)}, \tag{4.115}$$

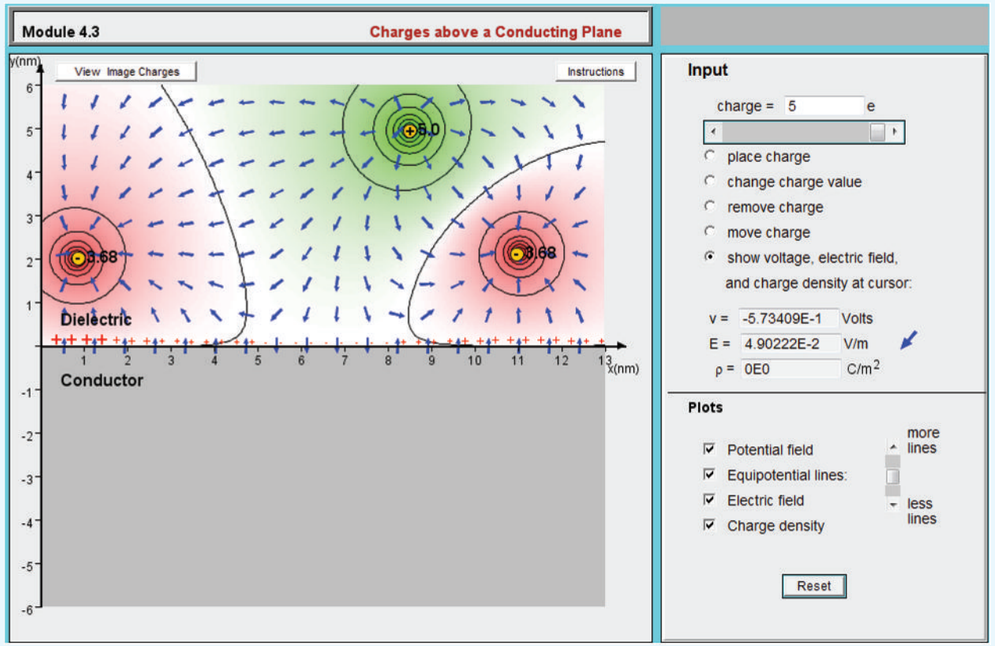
其中 V 是导体之间的电势（电压）差。电容以法拉 (F) 为单位，相当于库仑每伏 (C/V)。

导体表面自由电荷的存在会产生电场 E（图 4-25），其电场线始于正电荷，终止于负电荷。由于 E 的切向分量在导体表面总是消失，因此 E 始终垂直于导电表面。任一导体表面上任意点的 E 法向分量由下式给出

$$E_n = \hat{\mathbf{n}} \cdot \mathbf{E} = \frac{\rho_s}{\varepsilon}, \quad \textbf{(at conductor surface)} \tag{4.116}$$

其中 ρ_s 是该点的表面电荷密度，$\hat{\mathbf{n}}$ 是同一位置的向外法向单位矢量，ε 是分隔该点的介电介质的介电常数。

Module 4.3 Charges above a Conducting Plane When electric charges are placed in a dielectric medium adjoining a conducting plane, some of the conductor's electric charges move to its surface boundary, thereby satisfying the boundary conditions outlined in **Table 4-3**. This module displays **E** and V everywhere and ρ_s along the dielectric–conductor boundary.



conductors. The charge Q is equal to the integral of ρ_s over surface S (**Fig. 4-25**):

$$Q = \int_S \rho_s \, ds = \int_S \varepsilon \hat{\mathbf{n}} \cdot \mathbf{E} \, ds = \int_S \varepsilon \mathbf{E} \cdot d\mathbf{s}, \qquad (4.117)$$

where use was made of Eq. (4.116). The voltage V is related to **E** by Eq. (4.39):

$$V = V_{12} = -\int_{P_2}^{P_1} \mathbf{E} \cdot d\mathbf{l}, \qquad (4.118)$$

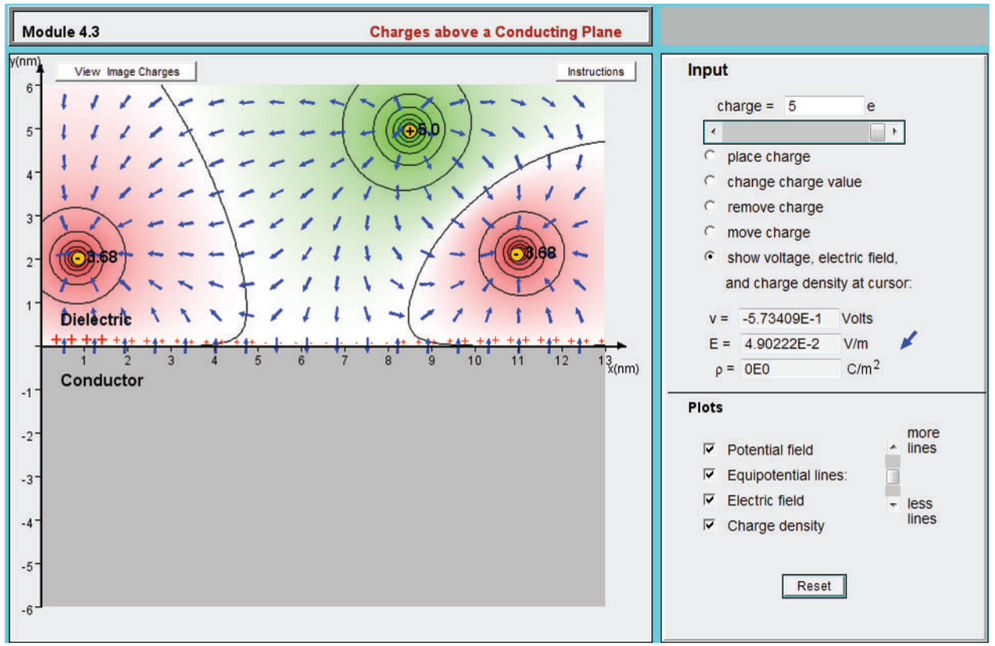
where points P_1 and P_2 are any two arbitrary points on conductors 1 and 2, respectively. Substituting Eqs. (4.117) and (4.118) into Eq. (4.115) gives

$$C = \frac{\int_S \varepsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}} \quad \text{(F)}, \qquad (4.119)$$

where l is the integration path from conductor 2 to conductor 1. To avoid making sign errors when applying Eq. (4.119), it is important to remember that surface S is the $+Q$ surface and P_1 is on S. [Alternatively, if you compute C and it comes out negative, just change its sign.] Because **E** appears in both the numerator and denominator of Eq. (4.119), *the value of C obtained for any specific capacitor configuration is always independent of* **E***'s magnitude.* In fact, C depends only on the capacitor geometry (sizes, shapes and relative positions of the two conductors) and the permittivity of the insulating material.

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模块 4.3 导电平面上方的电荷 当电荷放置在邻近导电平面的介电介质中时，导体的一些电荷会移动到其表面边界，从而满足表 4-3 中列出的边界条件。该模块随处显示 E 和 V，并沿电介质-导体边界显示 ρ_s。



导体。电荷 Q 等于表面 S 上 ρ_s 的积分（图 4-25）：

$$Q = \int_S \rho_s \, ds = \int_S \varepsilon \hat{\mathbf{n}} \cdot \mathbf{E} \, ds = \int_S \varepsilon \mathbf{E} \cdot d\mathbf{s}, \qquad (4.117)$$

其中使用了方程式。 （4.116）。电压 V 与 E 的关系如下：（4.39）：

$$V = V_{12} = -\int_{P_2}^{P_1} \mathbf{E} \cdot d\mathbf{l}, \qquad (4.118)$$

其中点 P_1 和 P_2 分别是导体 1 和 2 上的任意两个点。代入等式。 (4.117) 和 (4.118) 代入等式。 (4.115) 给出

$$C = \frac{\int_S \varepsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_l \mathbf{E} \cdot d\mathbf{l}} \quad \text{(F)}, \qquad (4.119)$$

其中 l 是从导体 2 到导体 1 的积分路径。为避免应用方程式时出现符号错误， （4.119），重要的是要记住曲面 S 是 +Q 曲面，P_1 在 S 上。[或者，如果计算 C 并且结果为负，只需更改其符号。]因为 E 出现在方程 1 的分子和分母中。 （4.119），对于任何特定电容器配置获得的 C 值始终与 E 的大小无关。事实上，C 仅取决于电容器的几何形状（两个导体的尺寸、形状和相对位置）和绝缘材料的介电常数。